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import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
X = np.array([
    [25, 30000, 2, 1],
    [30, 50000, 5, 0],
    [35, 60000, 7, 0],
    [22, 20000, 1, 2],
    [40, 80000, 10, 0],
    [28, 35000, 3, 1],
    [45, 90000, 15, 0],
    [24, 25000, 2, 2],
    [32, 55000, 6, 0],
    [38, 70000, 9, 0],
    [26, 28000, 2, 2],
    [42, 85000, 12, 0],
    [29, 45000, 4, 1],
    [36, 65000, 8, 0],
    [23, 18000, 1, 3],
    [31, 48000, 5, 1],
    [44, 78000, 11, 0],
    [27, 32000, 2, 2],
    [39, 72000, 10, 0],
    [33, 52000, 6, 1]
])

y = np.array([
    1, 1, 1, 0, 1,
    1, 1, 0, 1, 1,
    0, 1, 1, 1, 0,
    1, 1, 0, 1, 1
])

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

model = GaussianNB()

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model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("BANK LOAN PREDICTION USING NAIVE BAYES")
print("-----")
print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
new_customer = np.array([
    [30, 50000, 5, 0]
])
prediction = model.predict(new_customer)
probability = model.predict_proba(new_customer)
print("\nNew Customer Prediction:")
if prediction[0] == 1:
    print("Loan Approved")
else:
    print("Loan Not Approved")

print("\nPrediction Probability:")
print(probability)

```

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...  BANK LOAN PREDICTION USING NAIVE BAYES
...  -----

Actual Values:
[1 0 1 1]

Predicted Values:
[0 0 1 1]

Accuracy:
0.75

Confusion Matrix:
[[1 0]
 [1 2]]

Classification Report:

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		Classification Report:			
		precision	recall	f1-score	support
0	0.50	1.00	0.67	1	
1	1.00	0.67	0.80	3	
accuracy				0.75	4
macro avg		0.75	0.83	0.73	4
weighted avg		0.88	0.75	0.77	4

```

New Customer Prediction:
Loan Approved

Prediction Probability:
[[2.23652067e-12 1.00000000e+00]]

```